Qingyan (Selena) Lin

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EDUCATION

Georgia Institute of Technology, Atlanta, GA Expected Graduation: 05/2027

Bachelor of Science in Electrical Engineering, Concentration: Robotics and Electronic Devices; Minor: Industrial Design

Cumulative GPA: 3.65 (Dean's List)

Clubs: Electrical Lead, RoboCup RoboJackets – Competitive Robotics | Committee Member, Chinese Student Association

INTERNSHIPS

HSM Solutions, Hickory, NC | *Automation and Machine Vision Intern* 06/2026 - Present

- Developed and validated industrial machine vision inspection systems using LMI vision systems, conveyors, barcode decoders, and designed a custom control panel for vision inspection equipment, enabling future optimization.
- Designed and implemented automated inspection pipelines in GoPxL to verify the handrail's hole position and diameter, flat angle offset, bend angle, and surface defects, improving inspection accuracy and consistency.
- Configured PLC communication that integrated conveyor system, vision system, and pick-and-place machine to enable reliable automation workflows.
- Collaborated with engineering teams and vendors to troubleshoot issues, gather requirements, and deliver maintainable, production-ready automation solutions.

EXTRACURRICULARS AND PROJECTS

Georgia Institute of Technology, Atlanta, GA | *LED Brightness and Pattern Control Peripheral* 04/2025

- Developed an FPGA peripheral in VHDL featuring PWM control, clock division, register decoding, and FSM-based logic for configurable LED behavior
- Performed hardware validation, timing verification, and system debugging to ensure stable real-time performance and correct functionality on FPGA hardware

Georgia Institute of Technology, Atlanta, GA | *Breakout Arcade Game on Mbed Microcontroller* 03/2025

- Developed a Breakout game in C++, integrating real-time LCD graphics, joystick-based controls, and event-driven gameplay on an embedded platform to deliver responsive and interactive user experiences
- Designed object-oriented game architecture featuring collision detection, adaptive ball physics, and dynamic brick management using STL vectors, randomized launch angles, enabling efficient state handling and gameplay behavior

Cancer Neurobiology & Nanotechnology Lab, Atlanta, GA | *Research Assistant* 08/2025 - Present

- Enhanced MATLAB scripts to clean and analyze experimental data and generate automated plots
- Developed parametric 3D models in OpenSCAD driven by MATLAB outputs, enabling rapid design iteration and preparation for future 3D printing and prototyping
- Implemented and calibrated resistance measurement procedures and documented experimental methods for repeatability and validation

RoboJackets @Georgia Institute of Technology, Atlanta, GA | *Electrical Lead* – RoboCup Team 08/2024 - Present

- Led electrical hardware contribution for RoboCup 2025, placing 2nd in the Obstacle Avoidance Technical Challenge, 3rd in the Most Improved Team, and 4th overall in the competition
- Built and validated a custom high-voltage power and discharge system, combining an isolated flyback converter with an IGBT-based controlled discharge stage
- Created onboarding curriculum and mini project for new electrical team members, covering topics including KiCad, Git/Github, PCB design, soldering, datasheet interpretation, and system-level design of soccer-playing robots

Georgia Institute of Technology, Atlanta, GA | *Teaching Assistant*, *Intro to Python* 08/2024 - 05/2025

- Guided students through robotics programming projects, specifically focusing on path planning algorithms, leading to a significant improvement in student performance on related assignments
- Demonstrated strong communication and teaching skills by leading recitations for 50 students, mentoring 5 students one-on-one, and hosting office hours for a class of over 800 students
- Dedicated an average of 10 hours per week to write and grade homework and exam questions with other teaching assistants, demonstrating strong collaboration and cross-functional teamwork skills while maintaining a 3.86/4.0 GPA

TECHNICAL SKILLS

Programming: Python, C/C++, Matlab, Simulink, ROS, LaTeX, Assembly, VHDL, Git/Github, ssh, Linux

Hardware: Circuit Design, analog electronics, PCB design, Electrical Testing/Debugging, Schematics/layout, ESP32, STM32, FPGAs, Embedded Systems, firmware, UART, SPI, I2C

Instrumentation: Soldering, Digital Multimeter, Hot Air Gun, Oscilloscope, Function Generator, Power Supply, 3D printing, logic analyzers, Rhino, SketchUp

Tools: KiCAD, Intel Quartus Prime, Multisim, NI LabVIEW, Photoshop, OpenSCAD, LTspice, Inventor, Microsoft Office Suite, Visual Studio Code, AI, Rhino, AutoCAD, Bill of Materials

Certifications: AutoCAD Professional Level II (Autodesk 2021), 3ds Max Professional Level II (Autodesk 2021)

Coursework: Digital Design Lab, Circuit Analysis, Digital System Design, Digital Signal Processing, Feedback Control Systems, Microelectronic Circuit, Electromagnetic, Embedded Systems Design, Circuits&Electronics Lab, Autonomous Mobile Robots, Statistics